

RATCHA BURI AGROMET, Thailand

WMO: 484640

Lat: 13.488N

Lon: 99.796E

Elev: 30

StdP: 14.68

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	63.6	66.2	54.4	63.0	70.2	57.6	70.9	71.5	15.5	80.0	14.1	79.8	3.9	320	0.328

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.8	98.0	80.2	96.3	79.9	94.8	79.6	82.8	92.6	82.0	91.3	81.5	90.6	5.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.6	159.0	86.9	79.9	155.2	86.0	79.2	151.8	85.2	46.8	93.0	45.9	91.4	45.3	90.8	91.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	11.3	9.6	60.1	100.2	3.0	2.2	58.0	101.8	56.2	103.1	54.6	104.3	52.4	105.9		
(5)			56.7	86.3	3.9	3.3	53.8	88.7	51.5	90.6	49.3	92.5	46.5	94.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.9	78.5	81.8	84.4	86.8	86.2	84.7	83.7	83.8	83.5	82.5	81.0	78.5
	DBStd	3.69	3.94	2.81	3.19	2.66	2.29	1.81	1.79	1.54	1.72	2.03	2.63	3.97
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12022	883	890	1065	1104	1123	1041	1044	1049	1004	1007	929	883
	CDD65	6547	418	470	600	654	658	591	579	584	554	542	479	418
	CDH74	70666	3789	5042	7151	8302	7845	6751	6284	6363	5803	5380	4407	3549
	CDH80	29506	1457	2292	3486	4331	3698	2792	2401	2457	2138	1778	1445	1231
Wind	WSAvg	3.1	3.3	3.0	3.4	3.3	2.8	2.6	2.8	2.9	2.9	2.8	3.6	4.3
Precipitation	PrecAvg	95.1	0.7	0.8	2.1	3.5	11.1	14.8	17.0	18.2	15.2	9.2	2.0	0.6
	PrecMax	115.0	4.0	2.4	9.8	14.1	17.9	18.7	24.8	25.7	21.7	14.4	5.2	2.2
	PrecMin	72.8	0.1	0.0	0.1	0.6	3.9	9.1	10.7	11.9	9.9	3.9	0.0	0.0
	PrecStd	9.4	0.7	0.7	2.0	2.6	3.0	2.3	3.3	3.2	3.1	2.6	1.5	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.5	95.4	98.0	100.8	99.5	96.1	94.6	94.8	94.3	92.9	92.7	93.3
		MCWB	76.6	77.4	77.8	79.7	81.8	81.0	79.8	80.6	80.8	80.4	79.9	77.6
	2%	DB	91.7	94.0	96.1	98.8	97.1	94.1	92.7	92.6	92.4	91.0	90.7	91.2
		MCWB	76.2	77.5	78.2	80.4	81.1	80.5	79.3	79.8	80.0	80.3	79.1	77.1
	5%	DB	90.0	92.5	94.6	96.9	95.3	92.3	91.2	91.1	90.8	89.6	89.2	89.1
		MCWB	75.4	77.0	78.4	80.2	81.0	80.1	79.0	79.4	79.8	79.9	78.8	76.3
	10%	DB	87.7	90.9	93.3	95.0	93.3	90.6	89.6	89.5	89.2	88.1	87.3	86.7
		MCWB	74.3	76.2	78.5	80.1	80.6	79.7	78.7	79.0	79.5	79.6	78.1	75.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.6	80.7	81.9	83.7	84.2	82.7	81.7	82.2	82.5	82.2	81.5	79.8
		MCDB	89.7	92.6	91.8	95.2	95.2	92.1	90.3	91.0	90.4	89.8	89.2	88.4
	2%	WB	78.3	79.4	80.8	82.6	82.9	81.8	80.7	81.0	81.4	81.2	80.7	78.6
		MCDB	87.5	90.2	91.2	93.6	93.0	90.6	89.2	89.4	89.2	88.5	88.2	87.6
	5%	WB	77.3	78.6	80.0	81.7	81.9	81.2	80.0	80.4	80.7	80.6	80.0	77.7
		MCDB	85.7	88.7	90.0	92.2	91.4	89.3	87.9	88.5	88.4	87.6	86.8	86.0
	10%	WB	76.4	77.9	79.3	81.0	81.2	80.5	79.3	79.7	80.0	80.0	79.1	76.6
		MCDB	83.9	87.4	88.8	91.0	90.2	88.0	86.6	87.0	87.0	86.2	85.2	84.2
Mean Daily Temperature Range	5% DB	MDBR	18.0	19.3	18.4	17.8	15.4	13.2	12.6	13.0	12.8	11.6	12.6	15.4
		MCDBR	20.5	20.6	20.2	19.6	17.7	15.3	14.9	14.9	14.8	13.7	15.1	18.0
		MCWBR	6.9	6.5	5.9	5.6	5.2	4.8	4.6	4.7	5.1	4.9	5.3	5.9
	5% WB	MCDBR	17.5	19.0	19.0	18.4	16.5	14.0	13.7	14.0	13.9	12.8	13.5	15.9
		MCWBR	6.5	6.5	6.0	5.9	5.6	5.0	4.7	5.1	5.6	5.0	5.3	5.9
Clear-Sky Solar Irradiance	taub	0.539	0.593	0.621	0.552	0.477	0.445	0.441	0.442	0.461	0.572	0.534	0.521	
	taud	1.966	1.863	1.827	2.044	2.306	2.397	2.406	2.388	2.324	1.966	2.045	2.032	
	Ebn at Noon	242	235	232	247	262	268	270	272	269	237	241	243	
	Edh at Noon	57	65	69	56	42	38	38	39	42	58	52	52	
All-Sky Solar Radiation	RadAvg	1548	1702	1755	1895	1675	1538	1451	1466	1506	1417	1483	1490	
	RadStd	94	79	141	124	150	119	108	110	107	97	112	78	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	N/A	N/A	N/A	N/A	+0.32	+0.36	N/A	N/A	N/A	N/A

Nomenclature: See separate page